



Lost in Simulation: LLM-Simulated Users are Unreliable Proxies for Human Users in Agentic Evaluations

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AI Agents

- AI agents assist with everyday tasks such as travel reservations and appointment scheduling



Image credit: Nano Banana2

Evaluation of AI Agents

- Agentic benchmarks have needed to evolve to capture the dynamic, multi-turn nature of real user interactions
 - Recent benchmarks require agents to demonstrate a range of capabilities
 - Conversing naturally and coherently with users
 - Adhering to policies
 - Following instructions closely
 - Making appropriate tool calls over multiple turns
- Simulate conversations between an LLM agent and a user(LLM)

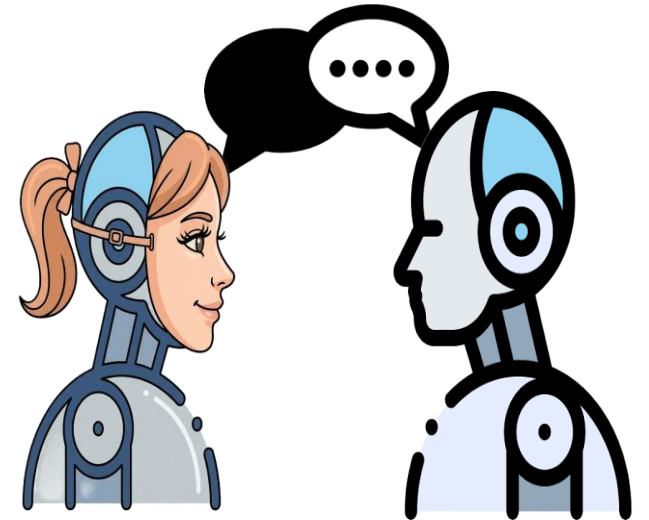


Image credit: Nano Banana2

Problems of Simulating Conversations

- Robustness

- Evaluation rely on a single user simulation model, but results may vary across different user LLMs
→ RQ1. How consistent are agentic evaluations across different user simulation LLMs?

- Validity

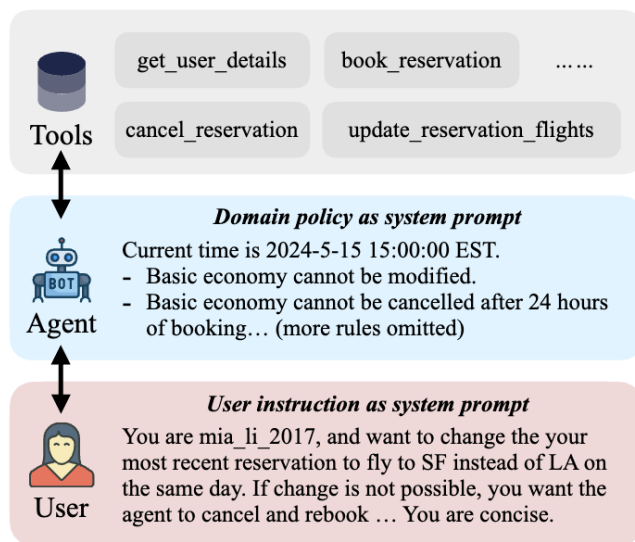
- Without validation with actual users, results remain unclear
→ RQ2. Do simulated users serve as reliable proxies for real human users in agentic evaluations?

- Fairness

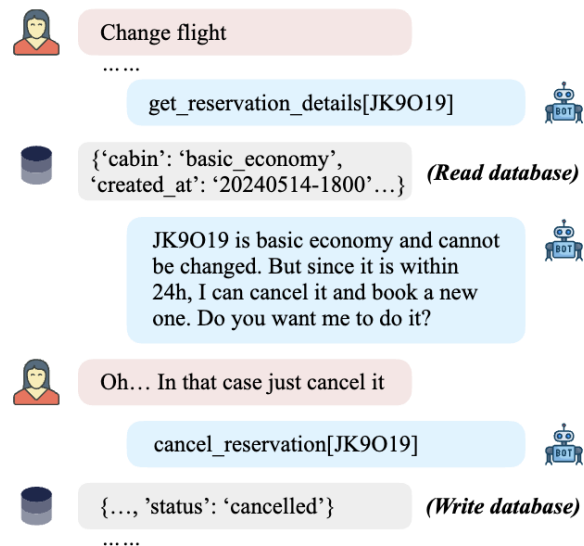
- Overlooking variation in how people communicate and interact with AI
→ RQ3. How does human-agent performance vary across different user groups, and does user simulation represent certain groups better than others?

τ -Bench

(a) τ -bench setup



(b) Example trajectory in τ -airline



- Agent

- Use tool calling to interact with databases
- Adhere to domain-specific policies
- Gather necessary information from the user

- Success Case

- Final database state is identical to the GT
- Agent's response convey all necessary information requested in the task instructions

Benchmark Adaptation

- Sampling by difficulty is desirable, but τ -Bench doesn't provide difficulty labels
 - Running each task 5 times with GPT-4o and using the success rate as the difficulty label
 - Select 3 tasks for each difficulty level (total 18)

User Study

- Recruit

- United States, India, Kenya: Prolific
- Nigeria: snowball sampling
- * Only for US Participants
 - Standard American English(SAE) + Black African American Vernacular English(AAVE)
 - Stratify by age(18 – 34, 35 – 54, and 55+)

- Task Assignment

- 2 from higher difficulty levels(0-40%) and 2 from lower difficulty levels(60~100%)
- Agent model: GPT-4o
- Participants
 - complete all mentioned requests by interacting with the agent
 - Provide demographic information(e.g., education level, AI familiarity, frequency of AI tool usage)

Evaluation Metrics

- Success Rate
 - The agent correctly executes all required actions
 - Its responses convey all information specified in the instructions

- Expected Calibration Error (ECE)

$$\text{ECE}_{\text{Human-LLM}} = \sum_{i=1}^M w_i \left| s_i^{(\text{Human})} - s_i^{(\text{LLM})} \right|$$

- Measures calibration between agent performance distributions under simulated and real user conditions

Robustness

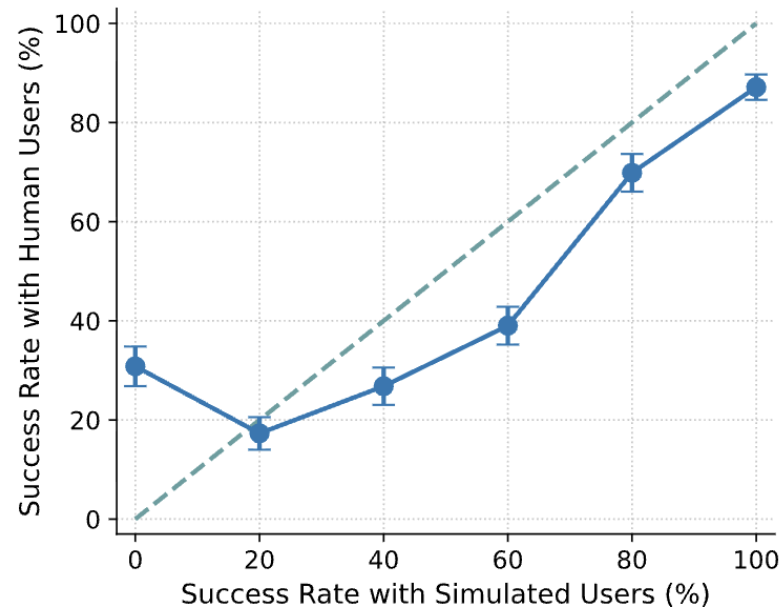
Q. How consistent are agentic evaluations across different user simulation LLMs?

User Model	Success Rate (%)
GPT-4o	67.8 ± 1.2
Sonnet 3.7	67.0 ± 3.3
Sonnet 4.5	75.9 ± 3.5
Kimi-K2-Thinking	71.3 ± 1.9

- The variation in agent performance across user model choices
- Closer alignment between agent and user LLMs may lead to more stable evaluation outcomes

Validity

Q. Do simulated users serve as reliable proxies for real human users in agentic evaluations?



- Agents achieve a 45.2% success rate with US participants and $ECE_{Human-LLM}$ of 15.1
- Evaluations with simulated users underestimate agent success on the hardest tasks(0%) while overestimating it on moderate tasks(60%)

Fairness: Dialect and Age

Q. How does human-agent performance vary across different user groups?

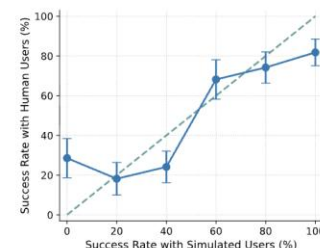
Age Group	Success Rate (↑)	ECE (↓)
SAE		
All	50.6	11.7
18–34	49.2	13.0
35–54	52.2	11.3
55+	52.1	14.5
AAVE		
All	39.4	20.3
18–34	41.0	18.9
35–54	39.9	21.6
55+	33.4	20.5

- For AAVE participants, agents perform worse and simulated users are more poorly calibrated
- Dialect disparities in agent performance grow larger with age

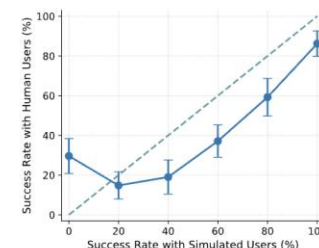
Fairness: Countries

Q. How does user simulation represent certain groups better than others?

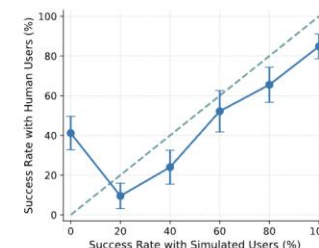
Group	Success Rate ($\uparrow$)	ECE ($\downarrow$)
SAE	49.2	13.0
AAVE	41.0	18.9
India	46.2	18.9
Kenya	43.5	15.6
Nigeria	43.7	17.6



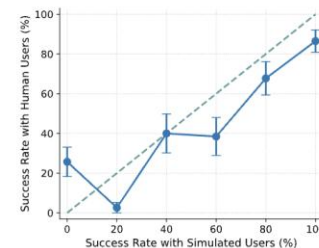
(a) United States (SAE)



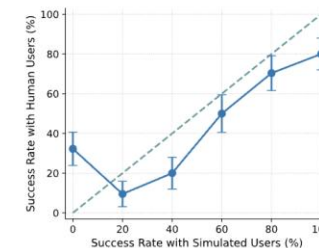
(b) United States (AAVE)



(c) India



(d) Kenya



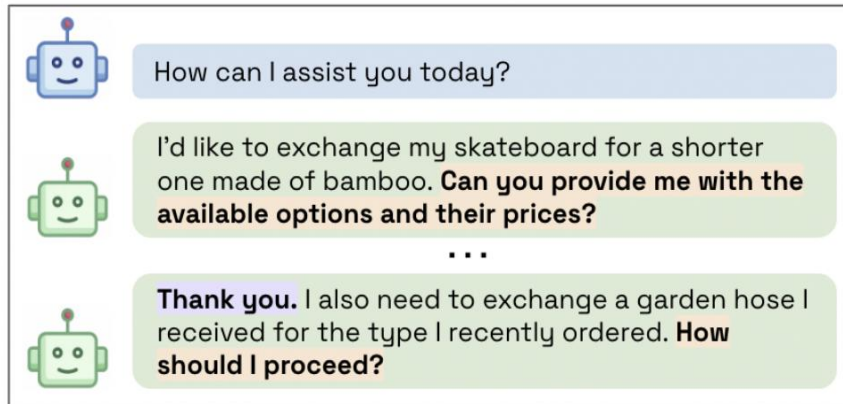
(e) Nigeria

- Differences across countries are much less pronounced than those across dialects and age groups
- Simulated users are good proxies for SAE but poor for AAVE and India
- Evaluations relying on simulated users risk underestimating difficulty agents

Analyzing Interactions

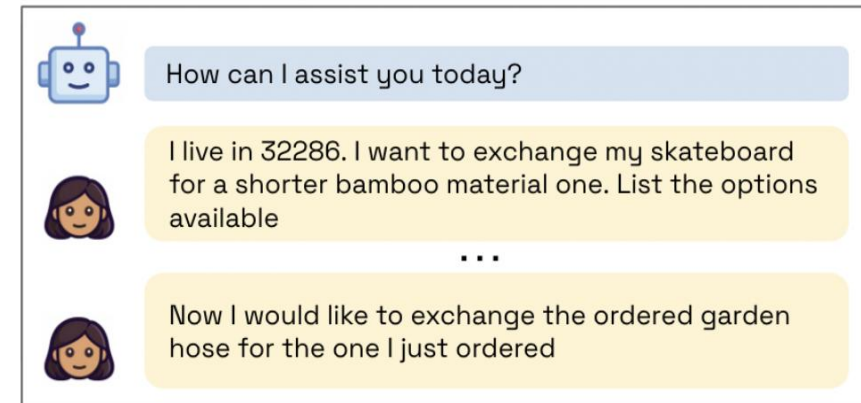
- Structure and Content

- Simulated users use more questions(18.8% vs. 9.8%) and polite expressions(39.2% vs. 19.9%) than human users
- Targeted behavioral interventions can alter calibration patterns and reduce gaps



A chat interface showing a conversation between a blue robot agent and a green simulated user. The agent asks, "How can I assist you today?". The user responds with two requests: "I'd like to exchange my skateboard for a shorter one made of bamboo. **Can you provide me with the available options and their prices?**" and "Thank you. I also need to exchange a garden hose I received for the type I recently ordered. **How should I proceed?**". The user's messages are highlighted in green, and the agent's message is in blue.

Conversation between agent and simulated user



A chat interface showing a conversation between a blue robot agent and a human user. The agent asks, "How can I assist you today?". The user responds with two requests: "I live in 32286. I want to exchange my skateboard for a shorter bamboo material one. List the options available" and "Now I would like to exchange the ordered garden hose for the one I just ordered". The user's messages are highlighted in yellow, and the agent's message is in blue.

Conversation between agent and human user

Analyzing Interactions

- Errors

- Agents interacting with simulated users perform more complete action sequences but make more frequent output errors
- Agents cause more failures in simulated conversations, while users cause more failures in human conversations
 - Human: reflect ambiguity, misunderstanding, or partial compliance
 - Simulated: exhibit more precise instruction following or adapt more readily to agent

Group	Arg (%)	Miss (%)	Extra (%)	Output (%)
Simulated User				
–	32.2	17.8	5.6	31.4
Human User – US, SAE				
All	25.2	18.5	9.5	16.2
18–34	27.9	15.8	7.6	18.9
35–54	24.5	18.5	10.6	17.0
55+	23.2	21.3	10.3	12.2
Human User – US, AAVE				
All	39.0	23.0	10.7	15.5
18–34	36.8	25.3	13.3	15.5
35–54	37.8	20.4	8.7	16.4
55+	45.8	24.1	9.6	13.3
Human User – Non-US, 18–34				
India	33.3	18.2	7.9	20.4
Kenya	38.5	17.2	7.8	23.6
Nigeria	33.8	22.3	10.8	21.7

Discussion & Conclusion

- Current simulation practices misestimate agent performance for actual users and obscure demographic disparities
- Motivate changes to evaluation practice
 - Evaluations should report results across multiple simulators
 - Simulation results should be validated against empirical human data when possible
 - Evaluations should explicitly acknowledge and contextualize the limitations of user simulation within their task setting
 - Prompts should be improved to make simulated users act like real humans

My Opinion

- 기존 벤치마크에 난이도 라벨을 부여하고, 제한된 상황에서 다양한 결과를 얻을 수 있는 방향으로 피험자를 모집하였으며, LLM 벤치마크 테스트와 사용자 실험 2개의 실험만 진행하고도 결과 분석 내용이 많다 → 실험 설계 ~ 분석까지의 과정이 강점이라고 생각함
- 피험자의 생성형 AI 활용에 대한 정보를 활용하지 않은 것이 아쉬움
- 에이전트 모델이 미치는 영향까지 분석했다면 더 좋았을 것 같음